

Lecture 6 Tutorial — Correlation and Regression

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Q1. Definitions (Fill in the blanks)

A correlation exists between two variables when the values of one variable are somehow _____ with the values of the other variable.

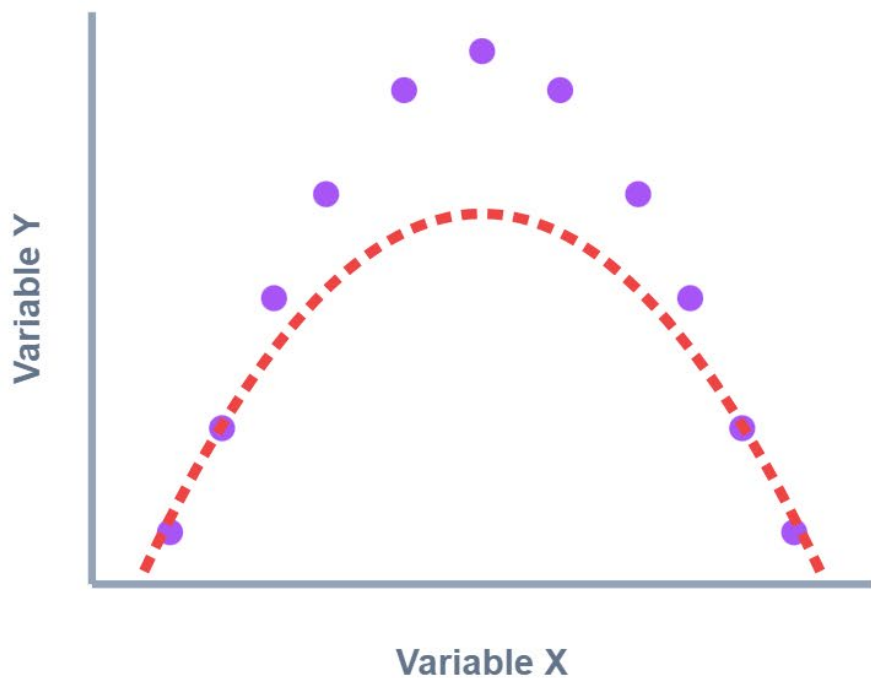
A linear correlation exists when the paired data can be approximated by a _____ line.

Understanding Correlation

Visualizing "Associated" vs. "Straight Line" relationships

1. General Correlation

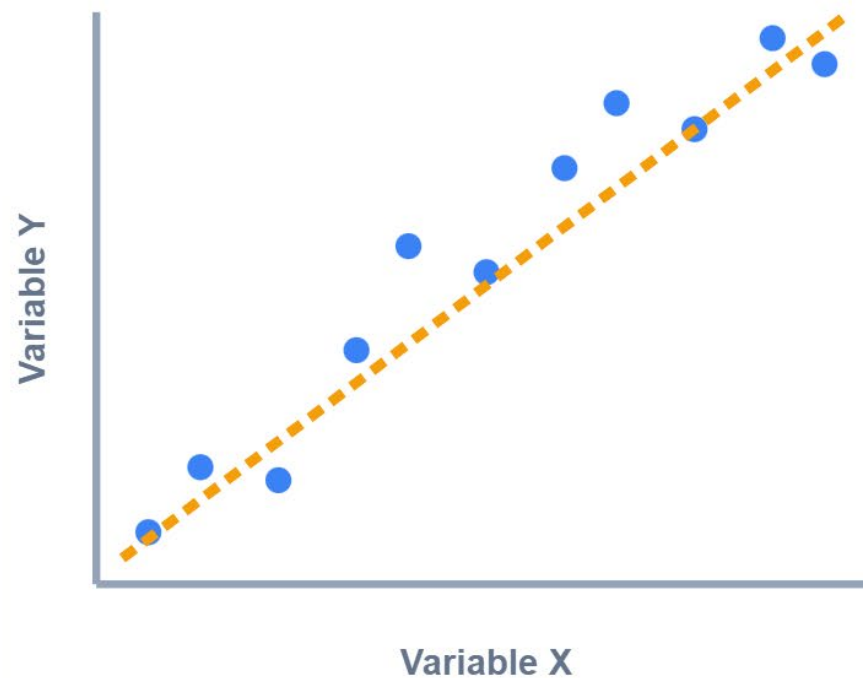
Variables are highly **associated**



Notice: They form a clear pattern (curve),
but a straight line would NOT fit here.

2. Linear Correlation

Approximated by a **straight** line



Notice: The points tightly hug a single
ruler-straight trajectory.

Q2. Computing from Summary Statistics (Calculation)

Given

$$n = 5, \quad \sum x = 10, \quad \sum y = 20, \quad \sum x^2 = 30, \quad \sum y^2 = 90, \quad \sum xy = 47.$$

1. Compute r (to 2 decimal places).
2. State whether the correlation is positive, negative, or zero.

Computing Pearson Correlation (r)

Breaking down the summary statistics into 3 manageable chunks

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2] \cdot [n \sum y^2 - (\sum y)^2]}}$$

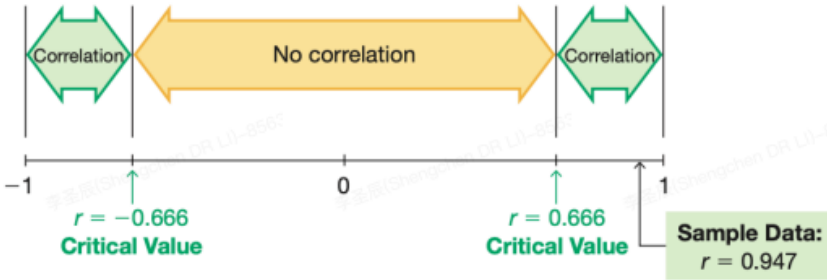


FIGURE 10-3 Critical r Values and the Computed r Value

Data: $n = 5$, $\sum x = 10$, $\sum y = 20$, $\sum x^2 = 30$, $\sum y^2 = 90$, $\sum xy = 47$

1. Numerator (Top)

$$\begin{aligned} n(\sum xy) - (\sum x)(\sum y) \\ = 5(47) - (10)(20) \\ = 35 \end{aligned}$$

2. Denominator (X Part)

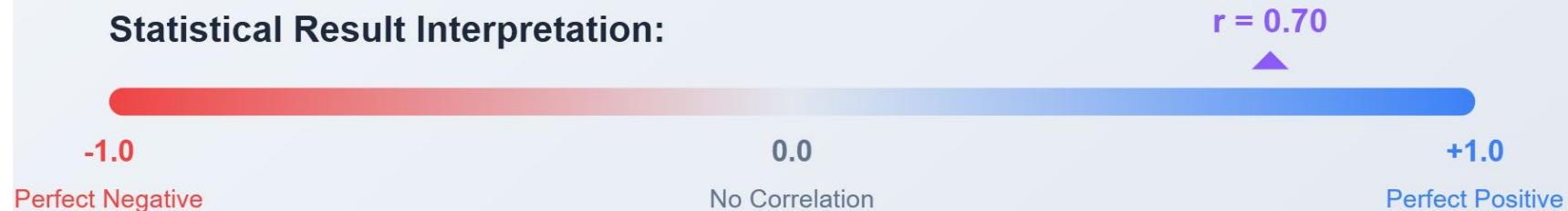
$$\begin{aligned} [n(\sum x^2) - (\sum x)^2] \\ = [5(30) - 10^2] \\ = 50 \end{aligned}$$

3. Denominator (Y Part)

$$\begin{aligned} [n(\sum y^2) - (\sum y)^2] \\ = [5(90) - 20^2] \\ = 50 \end{aligned}$$

$$r = \frac{35}{\sqrt{50 \times 50}} = \frac{35}{50} = 0.70$$

Statistical Result Interpretation:



Strong Positive Effect

Q3. Properties of r (True / False)

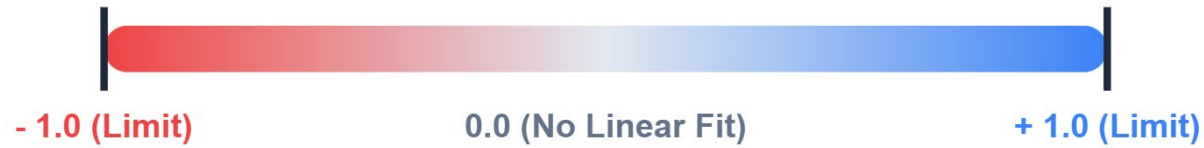
Write T or F.

1. The value of r is always between -1 and 1 (inclusive).
2. If all x values are converted to a different scale (e.g., multiplied by a constant), the value of r changes.
3. Interchanging all x values and y values changes the value of r .

1. "r is always between -1 and +1 (inclusive)."

✓ TRUE

r is standardized (Z-scores). It cannot escape the bounds of reality.

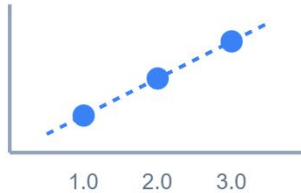


2. "If all X are scaled (multiplied), r changes."

✗ FALSE

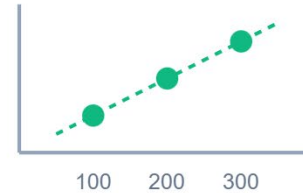
r is logically **Unitless / Dimensionless**. Scaling X does not change the core relationship.

X measured in Meters



EQUALS
=

X measured in Centimeters (x100)



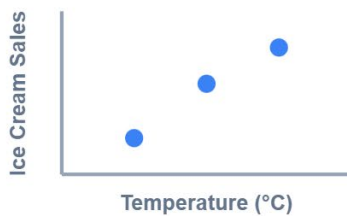
r = 0.95 is identical

3. "Interchanging x and y changes r."

✗ FALSE

r is **Symmetrical**. The relationship between X and Y is identical to the relationship between Y and X.

X-Y Relationship



EQUALS
=

Y-X Relationship (Swapped)



r = 0.85 is identical

Q4. What r Measures (Multiple Choice)

Choose the single best answer.

Which statement is most accurate?

- A. r measures the strength of any relationship, including nonlinear relationships.
- B. r measures the strength of a linear relationship, and it is not designed to measure nonlinear strength.
- C. r equals the slope of the regression line.
- D. r is unaffected by outliers.

Properties of the Linear Correlation Coefficient

1. The value of r is always between -1 and 1 inclusive. That is, $-1 \leq r \leq 1$.
 2. If all values of either variable are converted to a different scale, the value of r does not change.
 3. The value of r is not affected by the choice of x or y . Interchange all x values and y values, and the value of r will not change.
 4. r measures the strength of a linear relationship. It is not designed to measure the strength of a relationship that is not linear.
 5. r is very sensitive to outliers in the sense that a single outlier could dramatically affect its value.
-

Detailed Explanation: Core Properties of Pearson's Correlation Coefficient r



Core Concept of r

Quantifies the strength of linear relationships between continuous variables (Range: $[-1, 1]$).

- $r=1$: Perfect Positive | $r=-1$: Perfect Negative
- $r=0$: No linear correlation (not necessarily no relationship).



FINAL CONCLUSION

The only correct answer is
B



Teaching Analogy & Tips

- Ruler Analogy: r only measures "straight lines" (linear), not curves.
- Outliers: Like bent marks on a ruler—they heavily bias the result.
- Slope vs r : Slope is scale-dependent (gradient), r is standardized.



Option A: Measures any relationship (including nonlinear) [**INCORRECT**]

r is specifically designed for linear relationships only. It fails to capture nonlinear patterns (e.g., a perfect quadratic $y=x^2$ will yield $r \approx 0$).



Option B: Measures linear strength (ignores nonlinear) [**CORRECT**]

This aligns with the precise definition of Pearson's r . For nonlinear relationships, alternative statistics (e.g., Spearman's ρ) are required.



Option C: r equals the regression line slope (b) [**INCORRECT**]

They share the same sign but differ in value. Slope is scale-dependent while r is a scale-invariant, unitless measure of strength.



Option D: r is unaffected by outliers [**INCORRECT**]

r is extremely sensitive to outliers. A single extreme data point can drastically alter the value of r and even reverse the correlation direction.

Q5. Sensitivity to Outliers (Multiple Choice)

Choose the single best answer.

A single outlier can dramatically affect:

- A. only the sample mean, not r
- B. only the sample standard deviation, not r
- C. the value of r
- D. the value of r^2

Q5. Sensitivity to Outliers (Multiple Choice)

Question Statement

Choose the single best answer. Consider a random sample with a single extreme outlier:

"A single outlier can dramatically affect which measure?"

A. Only sample mean, not r (✗ Incorrect)

B. Only SD, not r (✗ Incorrect)

C. Value of r (✓ CORRECT)

Most direct & sensitive impact

D. Value of r^2 (⚠ Plausible)

Indirect consequence of r change



Core Concept: Outliers

An outlier is an observation that lies an abnormal distance from other values. It has a **disproportionately large effect** on many statistical measures, especially those based on squared deviations or linear relationships.



Analysis of Options

- A & B: Both wrong. Outliers affect mean/SD AND r significantly.
- C: Correct. Pearson's r is highly sensitive to outliers, which alter the regression line slope.
- D: Derivative. r^2 changes because r changes.



Final Conclusion

The single best answer is **C**. While outliers affect mean, SD, and r^2 , their most **direct and dramatic impact** in the context of correlation analysis is on the Pearson correlation coefficient (r).

Q6. Critical r and Significance (Decision)

A study reports a computed correlation

$$r = 0.947.$$

The critical values are

$$r = \pm 0.666.$$

1. Is the correlation statistically significant by the critical-value rule? (Yes / No)
2. Briefly justify using an inequality (one line).

Q6. Critical r & Significance

QUESTION CONTEXT

- Given Computed Correlation: $r = 0.947$
- Table Critical Values: $r = \pm 0.666$

Task: Determine significance & justify with an inequality.

CORE CONCEPT: THE CRITICAL-VALUE RULE

To test for significance (reject $H_0: \rho=0$), compare the absolute values. If the computed r is "far enough" from zero (greater than critical r), the correlation is significant.

$$| \text{Computed } r | > | \text{Critical } r |$$

STEP-BY-STEP ANALYSIS

STEP 1: Extract Values → Comp. $r = 0.947$; Crit. $r = 0.666$ (use absolute value).

STEP 2: Apply the Rule → Compare $|0.947|$ vs $|0.666|$.
Result: $0.947 > 0.666$ (TRUE)

STEP 3: Decision Logic → Since the rule holds true, we reject H_0 .

FINAL DECISION & JUSTIFICATION

1. Is the correlation significant? YES

2. Justification: $|0.947| > 0.666$

Q7. Spurious Correlation and Causality (Multiple Choice)

Choose the single best answer.

A high correlation value:

- A. proves x causes y
- B. proves y causes x
- C. does not by itself establish causality
- D. guarantees no confounding variables exist

Q7. Spurious Correlation and Causality (Multiple Choice)

Question: Choose the single best answer

A high correlation value indicates which of the following?

- A. proves x causes y
- B. proves y causes x
- C. does not by itself establish causality
- D. guarantees no confounders

Core Concept: Correlation vs. Causation

- Correlation: Two variables tend to change together.
- Causation: One variable directly causes change in another.
- Key Principle: Correlation does NOT imply causation.

Why High Correlation? (Possible Scenarios)

1. Direct Causation ($X \rightarrow Y$)
- 2. Reverse Causation ($Y \rightarrow X$)
3. Common Cause (Confounder Z) $\rightarrow$ Spurious Correlation

Analysis of Options

-  A: Proves X causes Y. Incorrect. Correlation alone cannot confirm direction.
-  B: Proves Y causes X. Incorrect. Reverse causation is also unproven by correlation.
-  C: Does not establish causality. Correct. This is the fundamental statistical rule.
-  D: No confounding variables. Incorrect. Confounders often create spurious correlations.

Final Conclusion: The single best answer is C.
Association (correlation) $\neq$ Causation; further investigation is always needed.

 **Classic Example: Ice Cream & Drowning**
Both rise in summer (Confounder Z: Hot Weather).
Correlation is spurious, not causal.

Q8. Regression Coefficients from r (Calculation)

Given

$$r = 0.60, \quad s_x = 2, \quad s_y = 5, \quad \bar{x} = 10, \quad \bar{y} = 100.$$

Use:

$$b_1 = r \frac{s_y}{s_x}, \quad b_0 = \bar{y} - b_1 \bar{x}$$

1. Compute b_1 .
2. Compute b_0 .
3. Write the regression equation in the form

$$\hat{y} = b_0 + b_1 x.$$

Q8. Regression Coefficients from r (Calculation)



GIVEN SUMMARY STATISTICS

Correlation (r) = 0.60 | Mean ($\bar{x}$)=10, ($\bar{y}$)=100
Std Dev (s_x)=2 | Std Dev (s_y)=5



CORE FORMULAS

Slope: $b_1 = r \times (s_y / s_x)$
Intercept: $b_0 = \bar{y} - (b_1 \times \bar{x})$
Equation: $\hat{y} = b_0 + b_1x$



REQUIRED TASKS

1. Calculate the slope (b_1) value
2. Calculate the intercept (b_0) value
3. Construct the final regression equation

STEP 1: FIND SLOPE (b_1)

$$\begin{aligned}b_1 &= 0.60 \times (5 / 2) \\b_1 &= 0.60 \times 2.5 \\b_1 &= 1.5\end{aligned}$$

Interpretation:
For each +1 unit in X , Y increases by 1.5 units.

STEP 2: FIND INTERCEPT (b_0)

$$\begin{aligned}b_0 &= 100 - (1.5 \times 10) \\b_0 &= 100 - 15 \\b_0 &= 85\end{aligned}$$

Interpretation:
Predicted value of Y when the variable X is 0.

STEP 3: BUILD EQUATION

$$\begin{aligned}\hat{y} &= b_0 + b_1x \\ \text{Substitute values:} \\ \hat{y} &= 85 + 1.5x\end{aligned}$$

Summary:
This linear equation models the predictive relationship between X and Y .

FINAL RESULTS

Slope (b_1): 1.5

Intercept (b_0): 85

Regression Eq:

$$\hat{y} = 85 + 1.5x$$

Q9. Prediction and Residual (Calculation)

Use the regression equation from Q8.

1. Compute the predicted value $\hat{y}$ when $x = 12$.
2. If the observed value is $y = 105$, compute the residual:

$$\text{Residual} = y - \hat{y}$$

3. State whether the point lies above or below the regression line.

Q9. Prediction and Residual (Calculation)

Question & Requirements

Use the regression equation from Q8: $\hat{y} = 85 + 1.5x$

1. Compute $\hat{y}$ when $x = 12$.
2. Compute residual ($y - \hat{y}$) if observed $y = 105$.
3. State if the point is above or below the regression line.

Core Concepts: Predicted Value & Residual

Predicted Value ($\hat{y}$): Estimated value on the regression line for a given x .

Residual: Vertical distance ($y - \hat{y}$) from the actual point to the line.

- **Positive:** Point lies **above** the line.
- **Negative:** Point lies **below** the line.

Step-by-Step Calculation

Step 1: Calculate Predicted Value

$$\hat{y} = 85 + 1.5(12) = 85 + 18 = 103$$

Step 2: Calculate Residual

$$\text{Residual} = y - \hat{y} = 105 - 103 = 2$$

Step 3: Determine Position

Residual is positive ($2 > 0$) → Point is **above** the line.

Final Results Summary

- Predicted Value ($\hat{y}$): 103
- Residual: 2
- Position: **Above** the regression line

Q10. Model Forms (Matching)

Match each regression type to its functional form.

Types:

- A. Linear
- B. Logarithmic
- C. Power
- D. Quadratic
- E. Exponential

Forms:

1. $y = a + bx$
2. $y = a + b \ln x$
3. $y = ax^b$
4. $y = ax^2 + bx + c$
5. $y = ab^x$

Write your answers as: A—____, B—____, C—____, D—____, E—____.

Q10. Model Forms (Matching)

Task: Match each regression type to its functional form.

Regression Types:

A. Linear B. Logarithmic
C. Power D. Quadratic
E. Exponential

Functional Forms:

1. $y = a + bx$ 2. $y = a + b \ln x$
3. $y = ax^b$ 4. $y = ax^2 + bx + c$
5. $y = ab^x$

Instruction: Write your answers in the format: A-_, B-_, C-_, D-_, E-_.
_.

💡 Core Concept: Model Selection

Different variable relationships demand specific mathematical models. Recognizing these functional forms (linear, logarithmic, power, quadratic, exponential) is fundamental to choosing the optimal regression model for accurate data fitting and interpretation.

A. Linear (1)

Standard straight-line relationship: $y = a + bx$.

B. Logarithmic (2)

y is linear to $\ln(x)$; shows decreasing marginal returns.

C. Power (3)

$y = ax^b$; linearized by log transformation (physics/allometry).

D. Quadratic (4)

Includes x^2 term; models parabolic relationships (peaks/troughs).

E. Exponential (5)

$y = ab^x$; models constant percentage growth/decay processes.

Final Answer A-1 B-2 C-3 D-4 E-5

Q11. Logistic Regression (Exact Values, No Calculator)

Logistic regression uses:

$$P(Y = 1) = \frac{1}{1 + e^{-z}}, \quad z = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

1. Compute $P(Y = 1)$ when $z = 0$.
2. Compute $P(Y = 1)$ when $z = \ln 3$. (*Hint: $e^{-\ln 3} = 1/3$*)

Using the decision rule “predict class 1 if $P(Y = 1) \geq 0.5$ ”, for which values of z is class 1 predicted?

- A. only $z > 0$
- B. $z \geq 0$
- C. only $z < 0$
- D. never (needs a different threshold)

Q11. Logistic Regression (Exact Values, No Calculator)

Question

Logistic regression uses: $P(Y=1) = 1/(1+e^{-z})$, where $z = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$

Tasks:

1. Compute $P(Y=1)$ when $z = 0$.
2. Compute $P(Y=1)$ when $z = \ln 3$ (Hint: $e^{-\ln 3} = 1/3$).
3. Predict class 1 if $P(Y=1) \geq 0.5$. Which z ? (A: $z > 0$ / B: $z \geq 0$ / C: $z < 0$ / D: Never)

Core Concept: Logistic Regression

Models binary outcome probability using the **sigmoid function** (maps z to $[0,1]$).

- At $z = 0$: $P(Y=1) = 0.5$ (midpoint)
- As $z \rightarrow +\infty$: $P(Y=1) \rightarrow 1$ (certainty)
- As $z \rightarrow -\infty$: $P(Y=1) \rightarrow 0$ (impossibility)

Step-by-Step Calculation

Step 1 ($z=0$): $1/(1+e^0) = 1/(1+1) = 1/2 = 0.5$

Step 2 ($z=\ln 3$): $1/(1+e^{-\ln 3}) = 1/(1+1/3) = 3/4 = 0.75$

Step 3 (Decision Rule): Sigmoid is increasing. $P \geq 0.5$ when $z \geq 0$. Matches **Option B**.

Final Answer Summary

1. Probability at $z=0$: 0.5
2. Probability at $z=\ln 3$: 0.75
3. Decision Rule: Option B ($z \geq 0$)